From Myth to Model:
Representation of “The Jew” In Generative AI

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Abstract

Antisemitism, a persistent form of cultural prejudice, is increasingly manifesting in subtle and modernized forms within contemporary society. This study examined how cultural representations of Jews are encoded in large language models (LLMs), which absorb societal discourse through training on vast corpora of human-generated text. To investigate these representations, we developed an indirect methodology: generating character biographies using Jewish and non-Jewish names, removing overt identity markers, and prompting the LLM to evaluate a range of psychological and sociocultural traits. Characters associated with Jewish (vs. non-Jewish) names were consistently rated as more competent, privileged, dominant, self-controlled, future-oriented, hierarchical, and obsessive—and as less likable and less collectivistic. When these trait profiles were mapped onto fictional characters, they aligned with archetypes historically associated with antisemitic tropes. These findings demonstrate how latent stereotypes can persist in AI systems despite explicit bias mitigation, highlighting the adaptability of prejudice and the structural embedding of cultural archetypes in contemporary technologies.

Keywords: Social Presentation, Generative AI, Jews, Antisemitism, Large Language Models

Introduction

Antisemitism, a persistent form of hatred targeting Jewish people, is experiencing a resurgence in contemporary society.[1] This alarming trend necessitates a comprehensive understanding of the evolving nature of antisemitism and its modern manifestations to effectively combat its impact and spread. At the core of antisemitism lies a socially constructed representation of "the Jew" (Smith, 1996). This representation, akin to other stereotypes extensively documented in social psychology and sociological research, profoundly shapes attitudes and actions toward Jewish individuals, playing a central role in their historical and ongoing persecution (e.g., Bilewicz et al., 2013).

Despite the variability observed across time and place, some features of the Jewish stereotype have demonstrated remarkable stability and continue to shape present-day manifestations of antisemitism (Johnson, 2023; Wodak, 2018). Central to the representation of "the Jew" is its construction as fundamentally and categorically *other*—distinct from and incompatible with the non-Jewish majority (Ascone, 2024). Antisemitic narratives frequently ascribe negative psychological and moral traits to Jews, particularly those associated with greed, cunning, and immorality. Common depictions include portrayals of Jews as avaricious and deceitful, motivated by an unrelenting pursuit of wealth (Bolton et al., 2024; Foxman, 2010; Shapiro, 1987), and as wielding disproportionate power and global influence (Becker, 2024). These longstanding tropes are often reinforced through conspiracy theories—such as the

[1] In 2024, the Anti-Defamation League (ADL) recorded 9,354 antisemitic incidents in the U.S.—a 5% increase from 2023 and an 893% increase over the past decade—averaging more than 25 incidents per day (ADL, 2024).

"Global Domination Conspiracy"—which claim that Jews covertly manipulate world events to serve their own interests at the expense of others (Fox & Topor, 2021).

In the post-Holocaust era, explicit expressions of antisemitic tropes were largely suppressed and deemed socially unacceptable in mainstream discourse. However, the rise of the internet and social media has enabled the renewed circulation of antisemitic content, often in subtle, coded, or suggestive forms (Becker et al., 2024; Finkelstein et al., 2018). Scholars have argued that such representations, even when actively disavowed, remain embedded within the cultural unconscious, shaping perception and communication in ways that are less overt but still consequential (Cheyette & Marcus, 1998). Research on implicit attitudes suggests stereotypes may operate below conscious awareness, influencing how individuals interpret the actions and intentions of individuals and collectives (Greenwald & Banaji, 1995). The persistence of these tropes poses risks not only at the interpersonal level but also within sociotechnical systems.

Such risks are especially evident in the case of large language models (LLMs)—a powerful class of generative artificial intelligence technologies trained on massive digital corpora. LLMs generate human-like text and are reshaping how language, knowledge, and social reasoning are processed and produced in contemporary society. They now support applications across sectors including customer service, education, healthcare, and legal support (Grossmann et al., 2023; Zhao et al., 2023). Beyond language generation, LLMs perform complex cognitive tasks with implications for labor, government, and industry. Empirical studies show that they can outperform humans in recommending socially appropriate behaviors, influence moral and interpersonal judgments, and produce persuasive political

messages that are often perceived as more logical and neutral than human-authored content (Bai et al., 2025; Mittelstädt et al., 2024; Pires et al., 2025).

Despite their impressive capabilities, LLMs raise significant ethical concerns due to their tendency to reproduce—and sometimes amplify—biases present in their training data. For instance, these systems have been shown to associate women with domestic roles or link African American names to negative traits (Bolukbasi et al., 2016; Caliskan et al., 2017; Meijer et al., 2024). These patterns are not merely theoretical: biased outputs can influence real-world decisions in domains like hiring, education, and loan approvals (Bowen et al., 2024; Salinas et al., 2023). The risks arise not because the models invent prejudice, but because they absorb and replicate the social inequalities embedded in human language. As such, LLMs do not simply reflect societal bias—they risk reinforcing and institutionalizing it under the appearance of algorithmic neutrality. Within this broader context, the reproduction of antisemitic representations—whether subtle or overt—should be examined as part of this wider pattern of encoded cultural assumptions.

Moreover, the well-documented replication of implicit biases in large language models makes the examination of their embedded representations an interesting venue for inquiry in its own right. Just as scholars have historically analyzed cultural artifacts—such as literature, visual media, and public discourse—to uncover the deep structure of antisemitic narratives, the outputs of LLMs, as derivatives of large-scale human linguistic activity, may offer a novel lens through which to investigate persistent societal biases. From this perspective, LLMs can serve not as direct reflections of society, but as heuristic instruments for generating hypotheses about the structure and transmission of cultural prejudice. This line of

inquiry resonates with structuralist and post-structuralist traditions that explore how language shapes and reveals the underlying logic of social systems (Barthes, 1972).

While studies on algorithmic bias in LLMs have not explicitly focused on anti-Jewish bias, scholarship in psychology and the social sciences has documented various dimensions of Jewish representation. For instance, research employing the Stereotype Content Model (SCM; Fiske et al., 2002) has identified an ambivalent representation of Jews.[2] Specifically, across studies, Jews are perceived as high in competence (e.g., smart, successful) but low in warmth (e.g., untrustworthy, immoral) (Bergsieker et al., 2012).

This pattern of ambivalence echoes broader findings on the duality of Jewish representations in antisemitism research. Despite the undeniable history of Jewish persecution—and their symbolic representation of the consequences of oppression—contemporary discourse often portrays Jews as a relatively privileged and successful group (Levine-Rasky, 2008; Slezkine, 2008). This duality is further reflected in their social positioning, casting Jews simultaneously as historical victims and as contemporary oppressors (Placzynta, 2024). Such paradoxical portrayals highlight the complexity of modern antisemitism. In light of this complexity, it is interesting to investigate whether the latent representation of Jews in LLMs aligns more closely with narratives of privilege or victimhood.

The present study systematically explored the encoded representations of Jews within large language models, with a focus on uncovering potential implicit biases. Specifically, we aimed to generate a relatively comprehensive representation of the

[2] See Cuddy et al. (2008) for a review.

psychological characterization of Jews—encompassing theoretically-grounded dimensions of social positionality and the Stereotype Content Model dimensions—alongside additional important attributes such as mental health, personality traits, and cultural values.

Method

Examining the latent representations of Jews in LLMs presents a unique challenge, particularly due to post-training processes designed to suppress culturally inappropriate or offensive responses. To overcome this obstacle, we employed a novel approach intended to identify underlying representations of “the Jew” by forming chains of associations that allow the LLM to reveal implicit biases. This methodology draws inspiration from projective psychological techniques and past work on AI bias (e.g., An et al., 2025; Armstrong et al., 2024; Salinas et al., 2024). By delving progressively deeper into the inferences generated through interactions with the LLM, we aimed to uncover both explicit and implicit encodings providing a more comprehensive understanding of antisemitic bias in these powerful language models. As LLMs become increasingly important targets of empirical investigation, methodological norms for interpreting and evaluating LLM-generated content are still emerging. Our approach reflects current best efforts to bring structure and theoretical grounding to this novel type of data, though we acknowledge that standards in this area are likely to evolve rapidly.

In this study, we focused on the outputs of OpenAI’s GPT model, specifically ChatGPT-4 Turbo, which was the most advanced and widely used AI model at the time of our analysis, with hundreds of millions of active users (Rudolph, 2023). GPT-4 was trained on a vast and diverse corpus of text, including books, websites,

academic articles, and other publicly available sources up to September 2021. This extensive training set enables the model to capture nuanced patterns in language, knowledge, and cultural discourse across a wide range of domains. To test the generalizability of our findings, the analyses were replicated using a powerful open-source model (DeepSeek). The investigation was conducted in independent steps (i.e., separate API prompt interactions) to ensure that the LLM would not base its response in later steps on previous interactions. Detailed prompting syntax and procedural steps are available in the Supplemental Materials.

Procedure

Step 1: Name Generation

GPT was instructed to create a list of names for Jewish and non-Jewish Americans, aged 18 to 80. The list included one male and one female name for each of the two categories, resulting in a total of 2 (ethnicity) × 2 (sex) × 63 (age) = 252 names (see Supplemental Materials for the Name Generation Prompt). Examples of Jewish names include Ethan Katz, Noah Weiss, and Gabriel Horowitz; non-Jewish examples include Tyler Johnson, Kyle White, and Dylan Wilson.

To validate that the names produced by GPT were representative of Jewish and non-Jewish individuals, we prompted a separate commercial LLM (Gemini) with the names generated by GPT and asked it to classify each name as either Jewish or non-Jewish and to rate its confidence. Gemini's classifications matched the GPT-generated labels in 97.6% of cases (i.e., 6 errors out of 252).

To assess how well GPT's generation of male and female names across different ages aligned with the actual demographic patterns of the U.S. population, we

used a widely adopted commercial tool built on large-scale name and age statistics (https://agify.io/). Results showed a good correlation between GPT-derived names for a given age and the tool's estimated average age for those names ($r = .51$, $p < .001$). For gender, GPT-generated names matched the expected label 95.6% of the time (i.e., 11 misclassified names out of 252).

Step 2: Biography Generation

For each of the 252 names, the LLM was prompted to write a short, 100-word biography. To ensure that the model projected characteristics onto the protagonist based on their name, we instructed it to take on the role of a novelist adept at selecting names that correspond with specific character traits. Biographies were required to include details such as the individual's place of residence, occupation, and both positive and negative personality traits. Biography generation prompts, along with an example pair of Jewish and non-Jewish biographies, can be found in the Supplemental Materials.

Step 3: Psychological Characteristic Ratings

In this step, all references to religion and the names themselves were removed from the biographies. GPT was then prompted to rate various psychological characteristics of the protagonists based on their biographies, using single-item questions and Likert scales. These trait inferences reflect the internal associations GPT has with the content it generated for Jewish exemplars. The rating generation prompt, along with an example pair of Jewish and non-Jewish character ratings, can be found in Table S1 of the Supplemental Materials.

Step 4: External Validation of GPT Ratings

DeepSeek Validation. To validate the ratings derived from the GPT-generated biographies, we conducted a follow-up analysis with DeepSeek (specifically, DeepSeek-V3-0324), a powerful open-source language model. In this step, DeepSeek was used solely to evaluate biographies generated by GPT. In Step 5, (described below) we conducted a full replication in which DeepSeek both generated and rated biographies independently.

Human Participant Ratings. To examine whether human participants perceived the biographies similarly to the LLM-generated ratings, we conducted a validation study in which participants rated the biographies on the dimensions associated with Jewish-derived exemplars in previous stages. Evaluations were collected via the Prolific online research platform. We recruited U.S. residents who identified English as their first language. After excluding participants who failed an attention check or were identified as outliers based on unusually short or long response times, the final sample included 378 U.S.-based participants (native English speakers; M_{age} = 39.5 years, *SD* = 13.32, range = 19–76). The sample included 166 male participants (43.9%) and 202 female participants (53.7%). Each participant rated one matched pair of biographies featuring a Jewish and a non-Jewish exemplar of the same gender and age. Each pair was evaluated by three independent participants.

To align the task with the nature of the LLM ratings, participants were instructed to estimate how *most people* would perceive the individual described in the biography, rather than report their personal views. Specifically, participants were told: "*Your task is to estimate how most people would rate the person described, rather than giving your own personal opinion. In other words, try to take the perspective of the average person and predict how the majority would perceive the individual in the*

biography." To encourage accurate estimation of perceived norms, we offered a monetary bonus to the top 10% of participants whose responses most closely matched the overall group average. This incentive technique, rooted in the Keynesian Beauty Contest framework, is commonly used in research on stereotypes and social norm perception (e.g., Fiske et al., 2002; Prentice & Miller, 1993).

Step 5: DeepSeek Replication

Finally, we conducted a full replication in which DeepSeek was used to generate and rate biographies independently, following the same procedure used for GPT.

The three-stage procedure with GPT was carried out in May 2024. The cross-platform validity studies—including the Gemini classification task, DeepSeek replication and validation, and human participant ratings—were conducted in June 2025.

Measures

We sought to assess the representation of Jewish versus non-Jewish exemplars across a range of psychologically meaningful dimensions. Each biography was evaluated by the LLM on the extent to which 84 characteristics applied to the individual described, using a 5-point Likert scale (1 = *strongly disagree*, 5 = *strongly agree*).

Stereotype Content Model Dimensions

Following the Stereotype Content Model (Fiske, 2018; Fiske et al., 2002), we measured stereotypes along two fundamental dimensions: warmth and competence.

These dimensions were assessed using 12 adjectives specifically selected to reflect stereotypical perceptions of warmth (e.g., friendly, likable) and competence (e.g., competent, effective).

Social Positioning

Perceived social positioning within societal power structures was measured using 10 adjectives reflecting social disadvantage and privilege. This list included terms such as victim, perpetrator, oppressed, and oppressor.

Superordinate Personality Traits

To measure personality traits, we used a modified version of the Big Five Inventory-2 (BFI-2; Soto & John, 2017). The original BFI-2 comprises 60 items across 15 facets and was optimized for psychometric validity in human participants. Some items were structured as direct opposites (e.g., "Thinks poetry and plays are boring" vs. "Is fascinated by art music or literature"), while others lacked clear counterparts (e.g., "Has a forgiving nature"). To adapt the measure for our purposes, we shortened the original list while ensuring that each trait included a reverse-coded item, to minimize assent bias in GPT's responses. We retained only items with a clear oppositional counterpart, selecting two items per personality dimension: one positively framed and one reverse-coded. The final set consisted of five adjective pairs, one for each Big Five trait.

For Openness, the items were "Thinks poetry and plays are boring" and "Is fascinated by art, music, or literature." For Conscientiousness, we used "Keeps things neat and tidy" and "Tends to be disorganized." For Extraversion, the pair included "Is dominant, acts as a leader" and "Prefers to have others take charge." For

Agreeableness, we selected "Is polite, courteous to others" and "Starts arguments with others." Finally, Neuroticism was measured using "Is moody, has up and down mood swings" and "Keeps their emotions under control."

Cultural Dimensions

To assess perceptions of the belief systems shaping the social context of each exemplar, we included a measure of cultural values. Both machine and human raters were asked to indicate the extent to which each belief was likely to characterize the individual described in the biography. Cultural values were measured using the Cultural Values Scale (Yoo et al., 2011), a 26-item instrument designed to measure Hofstede's cultural dimensions (2001; 2011) at the individual level. Responses were aggregated into five core dimensions: Power Distance, Collectivism, Masculinity, Uncertainty Avoidance, and Long-Term Orientation.

Psychopathology

Biases against social groups are sometimes expressed through the pathologization of their mental lives; indeed, antisemitic discourse has historically portrayed the psychological world of Jews as aberrant (see Discussion). To the best of our knowledge, no existing questionnaire systematically assesses the attribution of different psychopathologies to social targets. To address this gap, we compiled a list of items intended to capture such attributions. Specifically, we prompted GPT to "provide a list of prevalent but distinct mental health conditions." The resulting list consisted of 12 common mental health challenges discussed in popular culture (e.g., OCD, depression, substance abuse, autism spectrum). During the biography

evaluation phase, we prompted GPT to rate the extent to which each mental health condition described the individual.

Data Analysis

GPT and DeepSeek data were analyzed using a series of Bayesian paired samples *t*-tests, using the default prior on Cohen's *d* of Cauchy (0, 0.707) (Morey & Rouder, 2024). For each comparison, we report Bayes factors (BF_{10}), as well as the posterior medians and 95% credible intervals (CIs) of both the raw and standardized differences (Cohen's *d*). For the DeepSeek replication, we also computed a replication Bayes factor (Ly et al., 2019), which quantifies the relative support the DeepSeek data provide for the alternative versus the null hypothesis, given the observed GPT data. This approach was selected to account for the paired structure of the data—Jewish and non-Jewish biographies matched by age and gender—and to yield a continuous, probabilistic interpretation of the strength of evidence, which informed the subsequent analysis.

In the human validation study, which had a nested structure, we conducted a linear mixed-effects regression. Due to known challenges in specifying priors for mixed-effects models (van Doorn et al., 2023), we relied on maximum likelihood estimation. For each outcome, "type" (Jewish vs. non-Jewish) was included as a fixed effect, with the maximal random structure supported by the design (Barr et al., 2013): a random intercept and slope for "type" by biography "pair," and a random intercept for each participant nested within pair. We report the estimated mean difference, standardized estimated mean difference, and their 95% confidence intervals using Satterthwaite's method for estimating degrees of freedom (Kuznetsova et al., 2017). Finally, we fit a second model for each outcome excluding the fixed effect of "type,"

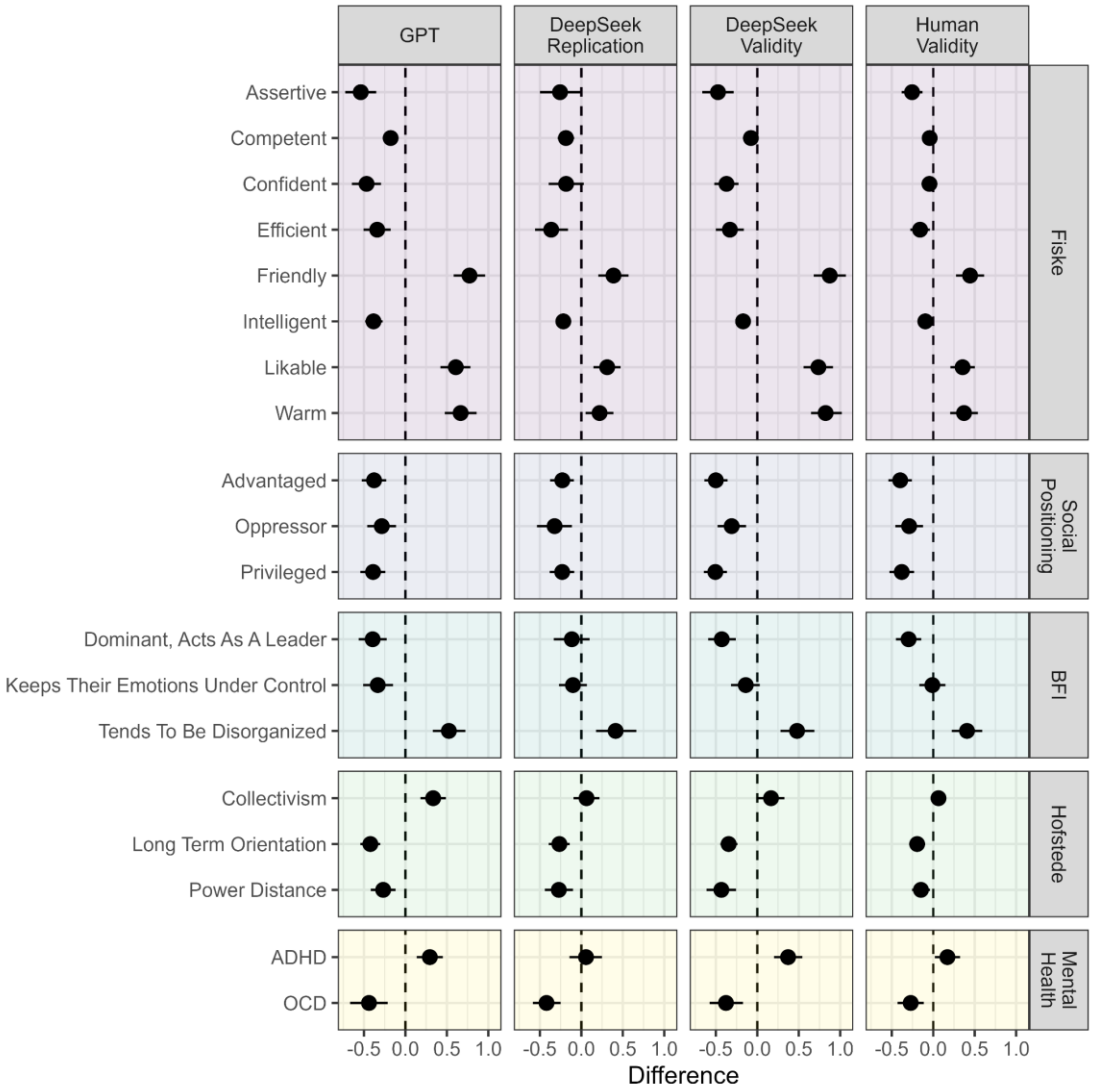
and used the BIC approximation to the Bayes factor (Wagenmakers, 2007) to test the effect of “type.” Assumptions of normality were assessed via visual inspection of Q–Q band plots (Loy et al., 2016; see Supplemental Materials). No substantial violations of normality were observed.

Results

The results of the Bayesian paired samples *t*-tests are presented in Supplemental Table S2. Group differences between Jewish and non-Jewish exemplars are further illustrated in Figure 1. Overall, the findings reveal consistent patterns aligned with stereotypical social representations of Jewish (vs. non-Jewish) individuals across multiple key dimensions.

Figure 1

Differences in Representation Between Jewish and Non-Jewish Exemplars



Note. Trait positioned further to the left are more strongly attributed to Jewish exemplars; traits positioned further to the right are more strongly attributed to non-Jewish exemplars.

Consistent with the SCM (Fiske et al., 2002), Jewish exemplars were rated higher on competence-related traits but lower on warmth-related traits compared to non-Jewish Americans. Specifically, Jews were rated as more intelligent (Cohen's d = -0.64, 95% CI [-0.49, -0.28], $BF_{10} = 3.90 \times 10^{8}$), confident (Cohen's d = -0.46, 95% CI [-0.64, -0.29], $BF_{10} = 2.39 \times 10^{4}$), assertive (Cohen's d = -0.51, 95% CI [-0.72, -0.35],

$BF_{10} = 2.02\times10^{5}$), competent (Cohen's d = -0.37, 95% CI [-0.26, -0.09], BF_{10} = 411.12), and efficient (Cohen's d = -0.37, 95% CI [-0.50, -0.18], BF_{10} = 375.76). However, they were also rated as less friendly (Cohen's d = 0.73, 95% CI [0.59, 0.96], $BF_{10} = 6.71\times10^{10}$), warm (Cohen's d = 0.63, 95% CI [0.48, 0.85], BF_{10} = 1.38×10^{8}), and likable (Cohen's d = 0.59, 95% CI [0.43, 0.79], $BF_{10} = 2.65\times10^{7}$).

With respect to social positioning, Jewish exemplars were rated as more privileged (Cohen's d = -0.45, 95% CI [-0.54, -0.24], $BF_{10} = 1.52\times10^{4}$), advantaged (Cohen's d = -0.44, 95% CI [-0.53, -0.23], $BF_{10} = 8.10\times10^{3}$), and oppressive (Cohen's d = -0.28, 95% CI [-0.46, -0.12], BF_{10} = 15.55), compared to non-Jewish counterparts.

In terms of personality, Jews were rated as more dominant and leader-like (Cohen's d = -0.41, 95% CI [-0.56, -0.22], $BF_{10} = 2.18\times10^{3}$), less disorganized (Cohen's d = 0.46, 95% CI [0.33, 0.73], $BF_{10} = 2.82\times10^{4}$), and more emotionally controlled (Cohen's d = -0.32, 95% CI [-0.51, -0.15], BF_{10} = 47.41), compared to non-Jewish Americans. Cultural value ratings revealed that Jewish exemplars were rated as less collectivistic (Cohen's d = 0.37, 95% CI [0.17, 0.49], BF_{10} = 428.75), more long-term oriented (Cohen's d = -0.62, 95% CI [-0.54, -0.30], $BF_{10} = 1.05\times10^{8}$), and more accepting of power distance (Cohen's d = -0.30, 95% CI [-0.42, -0.11], BF_{10} = 30.40). Regarding mental health conditions, Jewish exemplars were rated as more likely to have obsessive-compulsive tendencies (Cohen's d = -0.33, 95% CI [-0.67, -0.21], BF_{10} = 84.47) and fewer attention-deficit tendencies (Cohen's d = 0.33, 95% CI [0.14, 0.45], BF_{10} = 83.26).

DeepSeek and Human Validation of Biography Ratings

In these analyses, we assessed the reliability of the GPT-based ratings of GPT-generated biographies by comparing them to outputs from a second LLM (DeepSeek) and to human judgments. As shown in Figure 1 and detailed in Supplemental Table S4, DeepSeek ratings largely replicated the pattern of results observed with GPT. Jewish-associated exemplars were consistently rated as more intelligent, assertive, confident, and efficient; and less friendly, warm, and likable; more privileged, advantaged, and oppressive. They were also rated as more dominant, less disorganized, and as exhibiting greater long-term orientation and greater acceptance of power distance. Additionally, they received higher ratings for descriptions consistent with OCD and lower ratings for descriptions consistent with ADHD. We did not observe compelling evidence for differences in emotional control or collectivism.

Similarly, human ratings largely mirrored the GPT-based results (see Figure 1 and Supplemental Table S5). Jewish exemplars were rated as more intelligent, assertive, and efficient; and less friendly, warm, and likable; more privileged, advantaged, and oppressive. They were also perceived as more dominant, less disorganized, with greater long-term orientation and power distance acceptance. We did not observe compelling evidence for differences in competence, confidence, emotional control, or collectivism.

DeepSeek Replication Study

The validation studies described above relied on GPT-generated biographies. To examine the extent to which these Jewish-name-derived representations generalize to other LLMs, we conducted a replication study using DeepSeek. Specifically, we applied the same analytical approach used in the GPT study to analyze DeepSeek’s

own ratings of biographies generated by DeepSeek itself. Importantly, results were evaluated relative to the same prior benchmark values established in the original GPT analysis. The DeepSeek replication produced results that were largely consistent with the GPT findings. Of the 19 comparisons tested, 14 showed credible group differences (see Figure 1; Supplemental Table S3). Jewish exemplars were consistently rated as more intelligent, competent, and efficient; and less friendly and likable; more privileged, advantaged, and oppressive. They were also perceived as less disorganized, with greater long-term orientation, greater acceptance of power distance, and higher ratings for descriptions consistent with OCD. Small differences were observed for warmth and assertiveness, while no compelling evidence for group differences emerged for confidence, dominance, emotional control, or collectivism.

Qualitative Analyses: From Data to Archetype

The quantitative analyses identified 19 dimensions along which representations of Jewish and non-Jewish exemplars differed. To further concretize these patterns, we conducted a series of additional interactions using LLMs.

First, we prompted GPT to translate the 19-dimensional vector into a narrative description of a person, using the following instructions: "*These are characteristics of a person. The column standard deviations represent how many standard deviations does this person deviate from the population average. A positive number means that he has high levels of this attribute, and a negative number means that he has low levels of this attribute. Please translate the numbers to a comprehensive narrative description of the person that is sensitive to the degree to which the person exhibits each of these properties, as described in the table. Do not leave out any of the*

dimensions, but please be sensitive to the magnitude of the degree to which a person exhibits a specific characteristic."

Next, we asked GPT to identify fictional characters who matched the described profile, using the following prompt: "*Please generate a list of 20 fictional individuals from culture (e.g., film, television, video games, comic books, plays, novels, etc.) that fit this description well.*" The fictional characteristics most closely associated with the Jewish- and non-Jewish-derived vectors are presented in Table 1.

Table 1

Fictional Characters Most Similar to Jewish- and Non-Jewish-Derived Vectors

Non-Jewish Vector	**Jewish Vector**
Ron Weasley (*Harry Potter* series)	Sherlock Holmes
Chidi Anagonye (*The Good Place*)	Dr. Gregory House (*House, M.D.*)
Neville Longbottom (*Harry Potter* series)	Sheldon Cooper (*The Big Bang Theory*)
Will Byers (*Stranger Things*)	Tony Stark (*Iron Man* from Marvel Comics/Movies)
Samwise Gamgee (*The Lord of the Rings*)	Lisbeth Salander (*The Girl with the Dragon Tattoo series*)

Wendy Corduroy (*Gravity Falls*)	Walter White (*Breaking Bad*)
Ted Lasso (*Ted Lasso*)	Bruce Wayne (*Batman* from DC Comics/Movies)
Lenny Leonard (*The Simpsons*)	Michael Corleone (*The Godfather*)
Pam Beesly (*The Office*)	Light Yagami (*Death Note*)
Charlie Brown (*Peanuts*)	Dr. Manhattan (*Watchmen*)
Arthur Dent (*The Hitchhiker's Guide to the Galaxy*)	Hannibal Lecter (*The Silence of the Lambs*)
Luna Lovegood (*Harry Potter* series)	Frank Underwood (*House of Cards*)
Ben Wyatt (*Parks and Recreation*)	Spock (*Star Trek*)
Morty Smith (*Rick and Morty*)	Miranda Priestly (*The Devil Wears Prada*)
Glen Rhee (*The Walking Dead*)	Hannibal Lecter (*Red Dragon* / TV adaptations)
Bilbo Baggins (*The Hobbit*)	Tyrion Lannister (*Game of Thrones*)

Liesel Meminger (*The Book Thief*)	Dr. Strange (*Marvel* Comics/Movies)
Frodo Baggins (*The Lord of the Rings*)	Viktor (Victor) Frankenstein (*Frankenstein* by Mary Shelley)
Marge Simpson (*The Simpsons*)	Palpatine (Darth Sidious from *Star Wars*)
Ford Prefect (*The Hitchhiker's Guide to the Galaxy*)	Amy Dunne (*Gone Girl*)

To assess the consistency of these cultural archetypes across models, we replicated the process using DeepSeek. The resulting list of characters showed substantial overlap with that generated by GPT, with numerous characters appearing in both lists: Sherlock Holmes, Gregory House (*House M.D.*), Amy Dunne (*Gone Girl*), Walter White (*Breaking Bad*), Victor Frankenstein (*Frankenstein*), Tywin Lannister (*Game of Thrones*), Lisbeth Salander (*The Girl with the Dragon Tattoo*), Tony Stark (*Iron Man*), and Light Yagami (*Death Note*). In addition, DeepSeek identified several unique characters: Patrick Bateman (*American Psycho*), Thanos (*Marvel Cinematic Universe*), L (*Death Note*), Niles Crane (*Frasier*), Rick Sanchez (*Rick and Morty*), Cersei Lannister (*Game of Thrones*), Senator Armstrong (*Metal Gear Rising*), Lelouch vi Britannia (*Code Geass*), and Severus Snape (*Harry Potter*).

To identify the cultural archetype underlying the fictional characters generated by both GPT and DeepSeek, we asked each model to list the traits common to the subset of overlapping figures. Both models produced similar descriptions, which were

then analyzed by a third LLM (Gemini), which was tasked with identifying the shared descriptors across the GPT and DeepSeek outputs.

The composite archetype described by Gemini reflects themes consistent with classic antisemitic tropes: “These fictional characters consistently display several shared characteristics: they possess exceptional intelligence and strategic brilliance, often acting as master manipulators who meticulously plan intricate schemes; they exhibit profound moral ambiguity and antihero traits, frequently bending or breaking rules, laws, and ethical codes to achieve their goals; they tend towards social detachment and isolation, struggling with deep relationships and often appearing emotionally guarded or arrogant; they are defined by an intense drive and unwavering determination, pursuing personal missions or objectives with obsessive focus that often overrides conventional morality; and they operate with an ‘ends justify the means’ mentality, willing to inflict collateral damage or subordinate personal ethics to achieve their ultimate aims.”

To validate that these characteristics indeed reflect antisemitic tropes, we asked GPT and DeepSeek to evaluate which social groups are attributed with such characteristics, using the following prompt: “I am conducting research on prejudice. I will give you a description of a group of people. Which group is often attributed with such negative, hurtful prejudice? Clearly, we do not mean that these are the true characteristics of members of this group. Only that these are characteristics that are attributed to them by prejudiced individuals. please list five potential groups to which these descriptions may apply, in descending order of likelihood…” Both GPT and DeepSeek rated Jewish people as the most likely group. With DeepSeek noting that Jews are: "Historically and persistently targeted by antisemitic conspiracy theories

portraying them as hyper-intelligent, manipulative, secretive global puppet masters controlling finance, media, or governments ("The Protocols of the Elders of Zion" trope). The "ends justify the means" and isolation/superiority tropes are central to these harmful stereotypes.” GPT noted that: “The stereotype of Jews as highly intelligent but manipulative, secretive, power-seeking, and emotionally detached has been central to antisemitic propaganda for centuries. From Nazi-era caricatures to modern conspiracy theories (e.g., "globalist elites"), these harmful traits have been falsely attributed to Jewish individuals in a systematic way.”

Discussion

Our study reveals systematic patterns in how Jews are represented within LLM-generated content. Consistent with the Stereotype Content Model (SCM; Cuddy et al., 2008), Jewish-named characters were rated significantly higher on competence-related traits (e.g., intelligence, confidence, assertiveness) and significantly lower on warmth-related characteristics (e.g., friendliness, warmth, likability). In terms of social positioning, our analyses revealed that—despite their historical persecution—Jewish-named characters were rated more as oppressors than as oppressed, and as more privileged and advantaged. Additional personality analyses showed higher ratings for dominance and emotional control, and lower ratings for disorganization. These personality patterns were accompanied by lower ratings on collectivism, and higher ratings on both power distance and long-term orientation. In the domain of mental health, Jewish-named characters were rated higher for obsessive-compulsive tendencies and lower for attention deficit traits. All of these patterns—spanning traits, social positioning, personality, and mental health—were consistently replicated across models and human raters.

To investigate how these trait patterns might inform broader cultural narratives, we prompted the LLM to identify fictional characters who embody this constellation of traits. The resulting cultural archetype echoed several classic negative depictions of Jews—for example as “master manipulators who plan intricate schemes,” characterized by “social detachment and isolation,” and as “pursuing personal missions or objectives with obsessive focus that often overrides conventional morality.”

Our findings underscore the risks posed by latent biased representations within LLMs, particularly their potential to reproduce and entrench cultural stereotypes. Because these associations often remain dormant until triggered by specific prompts, their latency makes them difficult to detect or regulate, allowing biased patterns to persist beneath the surface of seemingly neutral outputs. Even without producing overtly prejudiced content, LLMs may encode constellations of socially charged traits—such as intelligence, emotional detachment, or perceived privilege—that reflect deeper cultural archetypes. While these traits may not coalesce into explicit stereotypes in typical model outputs, their latent presence suggests that such associations could still emerge under certain conditions. As LLMs become increasingly embedded in everyday applications, careful scrutiny is needed—not only of their explicit outputs, but also of the deeper representational logics they encode and perpetuate.

The Representation of Jews as Both Competent and Privileged

The SCM (Fiske et al., 2002) reconceptualized group stereotypes as varying along two key dimensions: warmth (perceived intent) and competence (perceived capability). These dimensions create four distinct quadrants of social perception:

admired groups (high warmth, high competence), pitied groups (high warmth, low competence), despised groups (low warmth, low competence), and envied groups (low warmth, high competence). This framework has demonstrated remarkable consistency across cultures and historical periods (Fiske, 2018).

Our analysis indicates that representations of Jews in LLM-generated content align with classic SCM findings (Fiske et al., 2002)—in which Jews are consistently stereotyped within the high-competence, low-warmth quadrant, alongside groups such as East Asians. In our data, biographies generated from Jewish names were rated consistently high on competence-related traits (e.g., intelligent, efficient, assertive) and notably low on warmth-related traits (e.g., friendly, likable).

A noteworthy pattern in the current findings is that these same biographies were also rated—by both humans and LLMs—as higher not only in competence (i.e., intelligent, competent, efficient), but also in *privilege* (i.e., advantaged, privileged). Privilege is defined as "an unearned advantage derived from one's group membership" (Phillips & Lowry, 2018). The term is typically applied to high-status groups, and the "unearned" aspect of this attribution implies that status is not necessarily driven by competence.

Previous SCM research suggests that people tend to perceive a reciprocal relationship between status and competence, rather than attributing the high status of certain groups to unearned advantages. That is, high-status groups are not typically viewed as undeserving. Fiske et al. (2002) argued that although it is theoretically conceivable for people to apply "sour grapes" reasoning—e.g., "the out-group may have high status, but they inherited it, lucked out, or cheated, so they do not deserve it, and they actually are stupid" (p. 880)—empirical evidence suggests that

"stereotypes function to justify the status quo" (ibid). Consequently, the perceived threat posed by high-status out-groups is more often managed by denying their warmth, rather than by denying their competence or framing their success as illegitimate.

From this perspective, the joint attribution of both competence and privilege to the Jewish-derived biographies is interesting. It may reflect a particular nuance of the Jewish stereotype: the idea that Jews "achieve their status both by being smart, but also by cheating." Alternatively, it may reflect a shifting cultural landscape in which system-justifying stereotypes ("they are successful because they are smart") co-exist with "sour grapes" narratives ("they are successful because they help each other get ahead").

More broadly, our methodological approach—examining chains of association—may simulate how complex societal representations form. The high-competence, low-warmth profile appears to interact with privilege attributions to generate broader narrative patterns, consistently producing themes of moral ambiguity and manipulation. This suggests that the SCM's foundational dimensions can function as building blocks which, when combined with additional attributes like privilege, give rise to more elaborate cultural narratives.

From Traits to Archetypes

Our first-order quantitative analysis identified a constellation of traits frequently associated with Jewish-named characters—some negative (e.g., unfriendly, oppressive), others positive (e.g., intelligent, competent). Building on this, our qualitative analysis revealed that this particular trait combination often maps onto

familiar cultural archetypes. Fictional figures such as Tywin Lannister (*Game of Thrones*) and Walter White (*Breaking Bad*) exemplify this pattern, sharing characteristics central to the antisemitic "puppet master" trope—isolated, powerful, obsessively focused, and morally ambiguous.

This analysis aligns with Roland Barthes's theory (1972) of cultural texts which posits that surface meanings carry implicit associations linked to deeper structures that shape interpretation. LLMs, trained on massive corpora of human-generated content, may have identified and encoded such cultural templates. In this case, traits that appear benign—or even admirable—in isolation can, through combination and context, reconstitute historical prejudices in subtler, more insidious forms.

LLMs function as word predictors. Based on sequence of words they calculate the probability distribution over possible next words (Jurafsky & Martin, 2023). Just as language models generate coherent text by predicting the next word based on learned patterns, societies may construct archetypes by layering associations drawn from shared cultural input. Through repetition and narration, some initially set of noticeable traits may coalesce into an archetype. Even when consciously rejected, such archetypes may unconsciously shape perception, contributing to persistent, implicit bias.

The Representation of Jews as Emblems of Modernity

The distinctive pattern we observed—higher ratings on competence and efficiency coupled with lower warmth; elevated scores on emotional control combined with reduced collectivism; and higher ratings on both power distance and

obsessive-compulsive traits—takes on deeper meaning when considered within the broader context of modernization and its discontents.

Modernization[3] has profoundly reshaped human subjectivity (e.g., Giddens, 1990). Rapid cultural change generates significant societal *anxiety*, particularly surrounding the dissolution of traditional social bonds and hierarchies. It has fundamentally transformed human relationships, replacing tight-knit communal bonds with impersonal, competitive interactions, often yielding a sense of *isolation*. In market-based systems, individuals may witness their social roles vanish and observe rapid social mobility, which can evoke *envy*. Moreover, success in the modern world increasingly demands abstract thought and rationality over immediate subjective experience—potentially fostering *emotional detachment*.

Jews have often been cast as symbolic representatives of these modernizing forces (Slezkine, 2008). Historical antisemitic discourse has frequently portrayed Jews as agents of disruption—undermining traditional order and social cohesion (Goldberg, 2017). Marx (1844) framed Jews as the originators of capitalist exploitation, while Wagner[4] (1850/2022) characterized them as destroyers of human emotion and spirituality. Antisemitic narratives have also projected the negative emotions associated with the modern predicament onto the minds of Jews themselves. Goebbels (1941), for instance, portrayed Jews as inherently *anxious*, possessing a natural instinct for sensing danger and a drive for self-preservation that enables their

[3] We use the term "modernization" to refer to an ongoing process, rather than a fixed historical period (e.g., Giddens, 1990; Inglehart & Welzel, 2004). It denotes the shift of societies towards urban, industrial, individualistic, market-based organization—rather than referencing a specific era such as the 17th to 20th centuries.

[4] See numerous examples in the Randall Bytwerk's German Propaganda Archive (https://research.calvin.edu/german-propaganda-archive; Bytwerk, 2012).

escape. Similarly, Esser (1939) described Jewry as "the embodiment of materialism, the epitome of sensuality, of *greed*." The trope of emotional inauthenticity also appears in Wagner's (1850) claim that "The Jew... has *no true passion* (Leidenschaft)," and in Oberlindober's (1939) assertion that "Jews are a foreign body in every people," emphasizing their supposed isolation from society.

Our findings suggest that this historical association between Jews and the ailments of modern subjectivity—documented in antisemitism research (e.g., Goldberg, 2017; Slezkine, 2008)—persists and may now be encoded in LLMs. To the extent that such associations remain entrenched, one might predict that increases in anti-modernization sentiment—such as backlash against the consequences of industrialization, capitalism, and technology—could co-occur with increases in antisemitic discourse.

Limitations and Future Directions

This study offers a detailed and novel view of latent Jewish representations in LLMs, but there is ample room to deepen and expand the analysis. Future work could refine the constructs used and adopt more granular measures to better capture the full complexity of group representations. For instance, our personality measures drew from a shortened version of the Big Five Inventory-2, with each trait represented by a single facet. Future studies could incorporate the full range of facets—or even finer-grained personality aspects—to capture more nuanced profiles. Similarly, we used Hofstede's cultural dimensions due to their theoretical relevance and prominence in LLM and computational research, but frameworks like Schwartz's (1992) value taxonomy may offer complementary insights. For psychopathology, we developed custom items, as no validated tool currently exists for assessing mental health

stereotypes across social groups. Creating such instruments—and distinguishing among more specific forms of psychopathology—would be a valuable extension. Finally, while our stereotype content ratings relied on the warmth–competence model, additional frameworks that distinguish between dimensions such as morality, ideological beliefs, and socioeconomic status could further enrich our understanding of how LLMs construct group representations (e.g., Abele et al., 2021). These extensions would not only deepen the current findings but also align future work with advances in personality theory, cultural psychology, and stereotype modeling.

A central limitation of this work is the challenge of studying antisemitic bias in systems that are continually evolving. Language models are frequently updated with new data and alignment protocols, making it difficult to track how bias manifests over time. This challenge is compounded by the fact that cultural stereotypes themselves are not static—they shift across historical contexts, making it hard to determine whether the patterns we observed reflect current attitudes, enduring tropes, or outdated narratives. One path forward would involve training models on decade-specific corpora to trace historical shifts in cultural representation. A complementary approach could assess how newer versions of the same model respond to identical prompts, revealing whether updates reduce, amplify, or reshape existing biases. Comparing these changes with how human participants respond to similar tasks over time would offer a fuller picture of how cultural representations evolve across both human and machine systems.

Another limitation is the study’s focus on English-language models and U.S.-based Jewish representations. As training data expands to include more non-English sources, examining how Jewish stereotypes vary across cultures and languages

(Varnum et al., 2024) would provide a valuable cross-cultural extension of our findings.

A fundamental question that remains unresolved is why stereotypes like the ones we observed emerge in language models at all. What kinds of prompts, training dynamics, or data distributions cause these representations to be encoded and surface? Do models generalize from specific examples, absorb broad patterns from distributional associations, or mirror subtle tropes embedded in genre, tone, or framing? Clarifying these pathways is critical for distinguishing surface-level bias from deeper structural encoding. Without a clearer understanding of these mechanisms, efforts to mitigate bias will remain reactive rather than preventive.

Future work should explore technical strategies that could shed light on these open questions. For instance, varying prompting approaches—along dimensions such as tone, directness, narrative framing, or abstraction—could help clarify when and why certain stereotypes appear. Investigating how model parameters like temperature (a variable which determines the degree to which the model produces the most predictable response) affect stereotype expression would also help identify conditions under which bias is more or less likely to surface. Prompt engineering and output conditioning may serve not only as mitigation tools, but also as diagnostic probes for uncovering latent model behavior. Other promising directions involve zooming in on model architecture and deployment context. Comparing “reasoning” models (such as GPT-4 Turbo or DeepSeek-R1) with more traditional LLMs may help clarify how different forms of inference mechanisms affect the reproduction of cultural stereotypes. Likewise, examining smaller, on-device models—especially given their

widespread use in mobile applications—could illuminate how computational constraints shape representational patterns in socially consequential contexts.

Another productive avenue for future research involves testing how explicit identity cues influence model behavior. While our current design removed such cues to probe latent associations, prompting LLMs with Jewish versus non-Jewish names could expose how surface-level identity markers shape outputs in applied domains such as hiring decisions, customer service replies, or biographical summaries. This approach would parallel audit-style studies used in social psychology and may help uncover both implicit and explicit forms of bias in real-world model interactions.

Ultimately, the persistence of cultural stereotypes in LLMs reflects a deeper structural challenge in AI development—one that endures despite mitigation efforts such as curated training data, output moderation, and human feedback. Because these models learn from historically biased corpora, even seemingly neutral prompts can produce harmful results. Reducing this harm without sacrificing cultural nuance, empathy, or historical understanding may require rethinking how models are built and fostering closer collaboration between computer science and the social sciences.

Conclusion

Our analysis reveals how ancient prejudices persist in modern technological systems through sophisticated patterns of trait association and cultural coding. These patterns suggest that addressing bias in AI systems requires attention not only to explicit stereotypes, but also to the subtle ways in which seemingly neutral traits combine to reproduce traditional prejudices. This understanding may inform future efforts to develop more equitable AI systems—while also highlighting the persistent

challenge of distinguishing beneficial pattern recognition from the perpetuation of harmful bias.

The transformation of historical antisemitic tropes into contemporary anxieties about modernization and social change—as reflected in our analysis of LLM representations—demonstrates both the adaptability of prejudice and the importance of historical context in understanding present-day bias. Our findings contribute to a deeper understanding of how cultural stereotypes persist and evolve, and they reinforce the need for continued research into mitigation strategies that address both explicit and implicit forms of prejudice.

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